Supplementary Table 12: Uptake rates (mmol gDW^{-1} h^{-1}) for dietary compounds implemented to simulate Western and high fiber diet.

Metabolite ID	Exchange reaction ID	Metabolite name	Western Diet	High fiber Diet
Sugars				
arab_L	EX_arab_L(e)	L-arabinose	0.17878295	0.04736842
cellb	EX_cellb(e)	Cellobiose	0.07449289	0.01973684
drib	EX_drib(e)	2-deoxy-D-ribose	0.17878295	0.04736842
fru	EX_fru(e)	D-Fructose	0.14898579	0.03947368
fuc_L	EX_fuc_L(e)	L-fucose	0.14898579	0.03947368
gal	EX_gal(e)	D-Galactose	0.14898579	0.03947368
glc_D	EX_glc(e)	D-glucose	0.14898579	0.03947368
glcn	EX_glcn(e)	D-gluconate	0.14898579	0.03947368
lcts	EX_lcts(e)	Lactose	0.07449289	0.01973684
malt	EX_malt(e)	Maltose	0.07449289	0.01973684
man	EX_man(e)	D-Mannose	0.14898579	0.03947368
melib	EX_melib(e)	Melibiose	0.07449289	0.01973684
mnl	EX_mnl(e)	D-Mannitol	0.14898579	0.03947368
oxa	EX_oxa(e)	Oxalate(2-)	0.44695737	0.11842105
rib_D	EX_rib_D(e)	D-ribose	0.17878295	0.04736842
rmn	EX_rmn(e)	L-Rhamnose	0.14898579	0.03947368
sucr	EX_sucr(e)	Sucrose	0.07449289	0.01973684
tre	EX_tre(e)	Trehalose	0.07449289	0.01973684
xyl_D	EX_xyl_D(e)	D-xylose	0.17878295	0.04736842
strch1	EX_strch1(e)	Starch	0.25733909	0.06818182
Fiber				
amylopect900	EX_amylopect900(e)	Amylopectin	0.00001567	0.00034722
amylose300	EX_amylose300(e)	Amylose	0.00004702	0.00104167
arabinan101	EX_arabinan101(e)	Arabinan	0.00016628	0.00368369
arabinogal	EX_arabinogal(e)	Larch arabinogalactan	0.00002191	0.00048550
arabinoxyl	EX_arabinoxyl(e)	Arabinoxylan	0.00030665	0.00679348
bglc	EX_bglc(e)	Beta-glucan	0.00000007	0.00000156
cellul	EX_cellul(e)	Cellulose	0.00002821	0.00062500
dextran40	EX_dextran40(e)	Dextran 40, 1,6-alpha-D-Glucan	0.00017632	0.00390625
galmannan	EX_galmannan(e)	Carob galactomannan	0.00001411	0.00031250
glcmannan	EX_glcmannan(e)	Konjac glucomannan	0.00003288	0.00072844
homogal	EX_homogal(e)	Homogalacturonan	0.00012823	0.00284091
inulin	EX_inulin(e)	Chicory inulin	0.00047019	0.01041667
kestopt	EX_kestopt(e)	Kestopentaose	0.00282117	0.06250000
levan1000	EX_levan1000(e)	Levan, 1000 fructose units	0.00001411	0.00031250

Metabolite ID	Exchange reaction ID	Metabolite name	Western Diet	High fiber Diet
lichn	EX_lichn(e)	Lichenin from Icelandic moss	0.00008298	0.00183824
lmn30	EX_lmn30(e)	Laminarin	0.00047019	0.01041667
pect	EX_pect(e)	Pectin	0.00003339	0.00073964
pullulan1200	EX_pullulan1200(e)	Pullulan	0.00001175	0.00026042
raffin	EX_raffin(e)	Raffinose	0.00470194	0.10416667
rhamnogalurI	EX_rhamnogalurI(e)	Potato rhamnogalacturonan I	0.00001449	0.00032106
rhamnogalurII	EX_rhamnogalurII(e)	Wine rhamnogalacturonan II	0.00026699	0.00591483
starch1200	EX_starch1200(e)	Resistant starch	0.00001175	0.00026042
xylan	EX_xylan(e)	Oat spelt xylan	0.00003206	0.00071023
xyluglc	EX_xyluglc(e)	Xyluglucan	0.00001315	0.00029124
Fat				
arachd	EX_arachd(e)	Arachidonate	0.00332813	0.00166406
chsterol	EX_chsterol(e)	Cholesterol	0.00495795	0.00247898
glyc	EX_glyc(e)	Glycerol	1.79965486	0.89982743
hdca	EX_hdca(e)	Hexadecanoate (n-C16:0)	0.39637090	0.19818545
hdcea	EX_hdcea(e)	Hexadecenoate (n-C16:1)	0.03651697	0.01825848
lnlc	EX_lnlc(e)	Linoleate	0.35910921	0.17955461
lnlnca	EX_lnlnca(e)	Alpha-linolenate	0.01756512	0.00878256
lnlncg	EX_lnlncg(e)	Gamma-linolenate	0.01756512	0.00878256
ocdca	EX_ocdca(e)	Octadecanoate (n-C18:0)	0.16928260	0.08464130
ocdcea	EX_ocdcea(e)	Octadecenoate (n-C18:1)	0.68144465	0.34072233
octa	EX_octa(e)	Octanoate (n-C8:0)	0.01294272	0.00647136
ttdca	EX_ttdca(e)	Tetradecanoate (n-C14:0)	0.06867567	0.03433784
Protein				
ala_L	EX_ala_L(e)	L-alanine	1	1
arg_L	EX_arg_L(e)	L-arginine	0.15	0.15
asn_L	EX_asn_L(e)	L-asparagine	0.225	0.225
asp_L	EX_asp_L(e)	L-aspartate	0.225	0.225
cys_L	EX_cys_L(e)	L-cysteine	1	1
gln_L	EX_gln_L(e)	L-glutamine	0.18	0.18
glu_L	EX_glu_L(e)	L-glutamate	0.18	0.18
gly	EX_gly(e)	Glycine	0.45	0.45
his_L	EX_his_L(e)	L-histidine	0.15	0.15
ile_L	EX_ile_L(e)	L-isoleucine	0.15	0.15
leu_L	EX_leu_L(e)	L-leucine	0.15	0.15
lys_L	EX_lys_L(e)	L-lysine	0.15	0.15
met_L	EX_met_L(e)	L-methionine	0.18	0.18

Metabolite ID	Exchange reaction ID	Metabolite name	Western Diet	High fiber Diet
phe_L	EX_phe_L(e)	L-phenylalanine	1	1
pro_L	EX_pro_L(e)	L-proline	0.18	0.18
ser_L	EX_ser_L(e)	L-serine	1	1
thr_L	EX_thr_L(e)	L-threonine	0.225	0.225
trp_L	EX_trp_L(e)	L-tryptophan	0.08181818	0.08181818
tyr_L	EX_tyr_L(e)	L-tyrosine	1	1
val_L	EX_val_L(e)	L-valine	0.18	0.18
Minerals, vitamins, other				
12dgr180	EX_12dgr180(e)	1,2-Diacyl-sn-glycerol (dioctadecanoyl, n-C18:0)	1	1
26dap_M	EX_26dap_M(e)	meso-2,6-Diaminoheptanedioate	1	1
2dmmq8	EX_2dmmq8(e)	2-Demethylmenaquinone 8	1	1
2obut	EX_2obut(e)	2-Oxobutanoate	1	1
3mop	EX_3mop(e)	3-methyl-2-oxopentanoate	1	1
4abz	EX_4abz(e)	4-Aminobenzoate	1	1
4hbz	EX_4hbz(e)	4-hydroxybenzoate	1	1
ac	EX_ac(e)	Acetate	1	1
acgam	EX_acgam(e)	N-acetyl-D-glucosamine	1	1
acmana	EX_acmana(e)	N-acetyl-D-mannosamine	1	1
acnam	EX_acnam(e)	N-acetylneuraminate	1	1
ade	EX_ade(e)	Adenine	1	1
adn	EX_adn(e)	Adenosine	1	1
adocbl	EX_adocbl(e)	Adenosylcobalamin	1	1
ala_D	EX_ala_D(e)	D-alanine	1	1
amp	EX_amp(e)	AMP	1	1
arab_D	EX_arab_D(e)	D-Arabinose	1	1
btn	EX_btn(e)	Biotin	1	1
ca2	EX_ca2(e)	Calcium(2+)	1	1
cbl1	EX_cbl1(e)	Cob(I)alamin	1	1
cgly	EX_cgly(e)	L-cysteinylglycine	1	1
chol	EX_chol(e)	Choline	1	1
chor	EX_chor(e)	Chorismate	1	1
cit	EX_cit(e)	Citrate	1	1
cl	EX_cl(e)	Chloride	1	1
cobalt2	EX_cobalt2(e)	Co2+	1	1
csn	EX_csn(e)	Cytosine	1	1
cu2	EX_cu2(e)	Cu2+	1	1

Metabolite ID	Exchange reaction ID	Metabolite name	Western Diet	High fiber Diet
dad_2	EX_dad_2(e)	2-deoxyadenosine	1	1
dcyt	EX_dcyt(e)	Deoxycytidine	1	1
ddca	EX_ddca(e)	Laurate	1	1
dgsn	EX_dgsn(e)	Deoxyguanosine	1	1
fe2	EX_fe2(e)	Fe2+	1	1
fe3	EX_fe3(e)	Fe3+	1	1
fe3dcit	EX_fe3dcit(e)	Fe(III)dicitrate	1	1
fald	EX_fald(e)	Formaldehyde	1	1
fol	EX_fol(e)	Folate	1	1
for	EX_for(e)	Formate	1	1
fum	EX_fum(e)	Fumarate	1	1
gam	EX_gam(e)	D-Glucosamine	1	1
glu_D	EX_glu_D(e)	D-Glutamate	1	1
glyc3p	EX_glyc3p(e)	Glycerol 3-phosphate	1	1
gthox	EX_gthox(e)	Oxidized glutathione	1	1
gthrd	EX_gthrd(e)	Reduced glutathione	1	1
gua	EX_gua(e)	Guanine	1	1
h	EX_h(e)	Proton	1	1
H2	EX_h2(e)	Hydrogen	1	1
h2o	EX_h2o(e)	Water	10	10
h2s	EX_h2s(e)	Hydrogen sulfide	1	1
hxan	EX_hxan(e)	Hypoxanthine	1	1
indole	EX_indole(e)	Indole	1	1
k	EX_k(e)	Potassium	1	1
lanost	EX_lanost(e)	lanosterol	1	1
meoh	EX_meoh(e)	Methanol	10	10
metsox_S_L	EX_metsox_S_L(e)	L-Methionine Sulfoxide	1	1
mg2	EX_mg2(e)	Magnesium	1	1
mn2	EX_mn2(e)	Mn2+	1	1
mobd	EX_mobd(e)	Molybdate	1	1
mqn7	EX_mqn7(e)	Menaquinone 7	1	1
mqn8	EX_mqn8(e)	Menaquinone 8	1	1
na1	EX_na1(e)	Sodium	1	1
nac	EX_nac(e)	Nicotinate	1	1
ncam	EX_ncam(e)	Nicotinamide	1	1
nmn	EX_nmn(e)	NMN	1	1
no2	EX_no2(e)	Nitrite	1	1
no3	EX_no3(e)	Nitrate	1	1
orn	EX_orn(e)	Ornithine	1	1
pheme	EX_pheme(e)	Protoheme	1	1

Metabolite ID	Exchange reaction ID	Metabolite name	Western Diet	High fiber Diet
pi	EX_pi(e)	Hydrogenphosphate	1	1
pime	EX_pime(e)	Pimelate	1	1
pnto_R	EX_pnto_R(e)	(R)-Pantothenate	1	1
ptrc	EX_ptrc(e)	Putrescine	1	1
pydam	EX_pydam(e)	Pyridoxamine	1	1
pydx	EX_pydx(e)	Pyridoxal	1	1
pydx5p	EX_pydx5p(e)	Pyridoxal 5-phosphate	1	1
pydxn	EX_pydxn(e)	Pyridoxine	1	1
q8	EX_q8(e)	Ubiquinone-8	1	1
ribflv	EX_ribflv(e)	Riboflavin	1	1
sel	EX_sel(e)	Selenate	1	1
sheme	EX_sheme(e)	Siroheme	1	1
so4	EX_so4(e)	Sulfate	1	1
spmd	EX_spmd(e)	Spermidine	1	1
thm	EX_thm(e)	Thiamin	1	1
thymd	EX_thymd(e)	Thymidine	1	1
ura	EX_ura(e)	Uracil	1	1
uri	EX_uri(e)	Uridine	1	1
xan	EX_xan(e)	Xanthine	1	1
zn2	EX_zn2(e)	Zinc	1	1